

- 8 In a power station generator, a large rectangular coil is rotating at 50 revolutions per second in a magnetic field of magnetic flux density of 0.29 T. The coil as shown in Fig. 8.1 has 38 turns each 2.0 m long and 1.2 m wide.

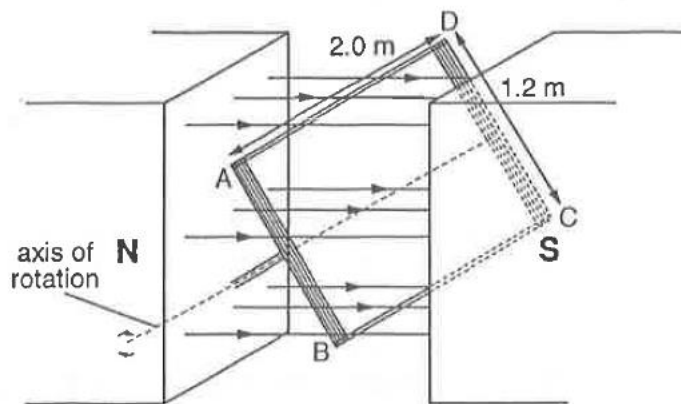


Fig. 8.1

The maximum output e.m.f. of the coil occurs when it moves near the plane of the magnetic field as shown in Fig. 8.2.

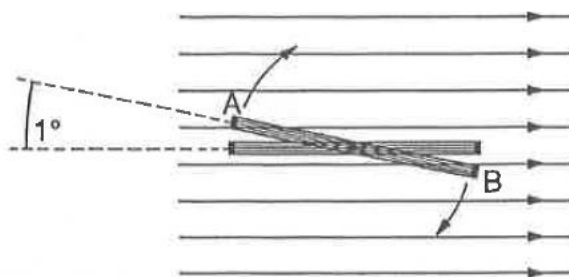


Fig. 8.2

(a) State Faraday's law of electromagnetic induction.

Faraday's Law states that the magnitude of the induced e.m.f ε in a conductor is directly proportional to the **rate of change of magnetic flux linkage** or to the **rate of cutting of magnetic flux**.

Many responses left out rate of cutting of magnetic flux

[1]

		(b)	For the coil moving through an angle of 1.0° near the plane of the magnetic field, calculate	
		(i)	the time taken for it to rotate 1.0°	
			time =s	[1]
			period = $1/50$ Time = $1/50/360 = 5.56 \times 10^{-5} \text{ s}$ <i>This was generally well done.</i>	
		(ii)	the flux cut by one turn of the coil in this time	
			flux cut =Wb	[3]
			$\phi = BA \cos\theta$ $= 0.29 \times 2 \times 1.2 \times (\cos 89^\circ - \cos 90^\circ)$ $= 0.0121 \text{ Wb}$ <i>Many responses could not get the correct component of the magnetic flux density, often using $\cos 1^\circ$ or did not note that there is a change of flux.</i>	
		(iii)	the e.m.f. generated by one turn of the coil in this time	
			e.m.f. =V	[2]
			$E = -d\phi/dt$ $= 0.0121 / 5.56 \times 10^{-5}$ $= 218 \text{ V}$ <i>Most could do this even with error carried forward.</i>	
		(iv)	the e.m.f. generated by all 38 turns of the coil in this time	

			e.m.f. =V	[1]
			218 x 38 = 8.27 kV	
			Most could do this even with error carried forward.	
	(c)	(i)	The value obtained in (b)(iv) is the peak value of the sinusoidal output of the coil. Calculate the r.m.s. value of the output of the coil.	
			r.m.s. value of the output =V	[1]
			$V_{\text{rms}} = 8269 / \sqrt{2} = 5.85 \text{ kV}$	
			Most could do this even with error carried forward.	
		(ii)	State the direction of the current induced inside AD as a result of this e.m.f. Explain how you deduced your answer.	
				
				
				[2]
			By Lenz's Law, to counter the increase in flux, must produce a field opposite to original on the coil and thus by Right Hand Grip rule, Current is from A to D. OR Use Fleming's Right Hand Rule with appropriate description in directions of the 3 fingers. Current is from A to D. <i>Several responses did not describe the directions in using Fleming's Right Hand Rule.</i>	
	(d)	(i)	State Lenz's law.	
				
				[1]
			Lenz's Law states that the <u>direction</u> of the <u>induced e.m.f</u> is such that it <u>produces effects</u> that <u>opposes the change that causes it</u> .	
			<i>Many responses talked about induced current, even though induced current may not always occur. Also many responses like to talk about the induced emf or current opposing the change rather than produce effects to oppose the change.</i>	
		(ii)	Explain the following two situations, using the laws of electromagnetic induction:	

1. A copper disc spins freely between the poles of an unconnected electromagnet as shown in Fig 8.3.

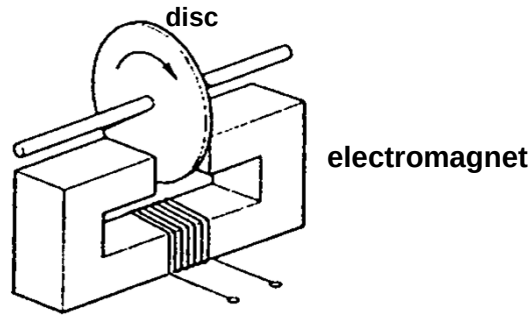


Fig 8.3

Describe and explain what will happen to the speed of rotation of the disc when a direct current is switched on in the electromagnet.

[4]

When the electromagnet is switched on, there is a magnetic field between the poles.

Thus the spinning disc cuts through the magnetic flux lines.

The cutting of flux lines results in an induced emf, by Faraday's Law.

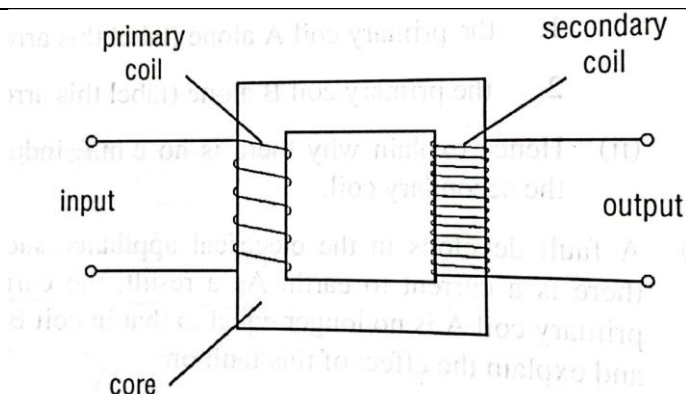
The disc is a closed circuit and thus current flows in it, which by Lenz's Law, acts in a direction that opposes the change that causes it in the first place.

The induced (eddy) currents in the spinning disc results in heat dissipation and energy loss, and hence this slows down the rotation of the spinning disc.

Many responses talked about change in magnetic flux. This is an inappropriate analysis of Faraday's disc.

Several responses also thought that the disc will eventually spin at a constant, albeit, lower rate, indicating their lack of understanding of the context.

2. A simple iron-cored transformer is shown below.



Suggest why the input voltage and the output e.m.f. have the same frequency.

[3]

Since magnetic flux density (B) is proportional to the current (I) flowing through the coil, B and I have the same frequency.

The B produced from the primary coil is linked to the secondary coil via the soft iron core. By Faraday's law, there will be an induced emf in the secondary coil given by $\frac{dB}{dt}$.

Hence, the induced emf will have the same frequency as the B in the primary coil, which is then has the same frequency as the I in the primary coil.

The crux of this question is to use laws of electromagnetic induction. Responses that do not use these laws will not get credit.

Many responses lack a logical flow. The first step is to establish that the flux density and hence the flux is proportional to the current hence the flux must change at the same frequency as the input voltage.

Next by Faraday's law, the induced emf is $\frac{dB}{dt}$ which means the secondary voltage has the same frequency with B.

Finally using the two previous statements, one can conclude the output voltage is at the same frequency as the input voltage.

Most responses make quick conclusions without establishing the link between the different quantities.

[Total: 20]

9 Some elements that are normally stable, such as lead (Pb), have isotopes which are radioactive. The